FLOODGAUGE · TRACEABILITY MATRIX

FloodGauge traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/03b-traceability.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Generated by .kit/trace.py

FloodGauge traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 19 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-009	PR-001	Street depth range and mounting height	FLG-CAL-001, FLG-BLD-001	Range calculation against sensor data (CAL), TRL 3	Met on paper: reference lens at 4.6 m clears a 4.3 m vehicle; range 1.9 to 5.0 m over the band, radar to 20 m
UR-001, UR-003	PR-002	Street depth accuracy	FLG-CAL-001, FLG-BLD-001, FLG-STM-001, FLG-OPT-001	Error budget (CAL), TRL 3; bench test over a tank at TRL 4	Met on paper with radar: 6.5 mm root sum square (12.0 mm worst-case sum), module accuracy assumed. **Not met** by the ultrasonic variant (27.0 mm compensated at 4.6 m)
UR-002	PR-003	Catch basin level range	FLG-CAL-001, FLG-BLD-001	Geometry of stilling tube and head (CAL), TRL 3	Met on paper: continuous from 1,250 to 330 mm below the road; wet probe trips at 310 mm
UR-002, UR-003	PR-004	Drain level accuracy	FLG-CAL-001, FLG-BLD-001	Error budget (CAL), TRL 3	Met on paper: 11.0 mm with the NTC in the head (19.7 mm without)
UR-005	PR-005	Sampling rate	FLG-CAL-001	Sampling logic review (CAL), TRL 3	Met by design
UR-004	PR-006	Alert latency	FLG-CAL-001, FLG-STM-001	Latency simulation (CAL), TRL 3	Met on paper at SF9 and SF10 on a private gateway: 82 s at the 95th percentile (85 s on a city server); 110 s worst case without a lost packet. Not met at SF11 (123 s) and SF12 (201 s) in event mode (independent review)
UR-013	PR-007	Radio airtime within network rules	FLG-CAL-001, FLG-STM-001	Airtime calculation (CAL), TRL 3	At risk:** normal mode 23.7 s/day at SF9; event mode within 1 % at SF7 to SF10 but not at SF11 or SF12 unless reporting stretches to 99 s or 181 s; normal mode exceeds fair use at SF10 and slower
UR-006	PR-008	Energy autonomy	FLG-CAL-001	Power budget against FieldNode (CAL), TRL 3	Met on paper: 68 days nominal, 47 days at -20 °C; 9.3 % of the FieldNode 100 mW allowance
UR-007	PR-009	Survive submersion of the drain head	FLG-CAL-001, FLG-STM-001, FLG-OPT-001	Enclosure design review (CAL), TRL 3; immersion test at TRL 4	At risk:** electronics potted, but the transducer face keeps the module's IP67 seal
UR-011	PR-010	Reject false readings	FLG-CAL-001, FLG-OPT-001	Plausibility logic review (CAL), TRL 3; field log at TRL 4	To verify at TRL 4: rules defined, including a recorded dry-road background against the curb echo

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-012	PR-011	Privacy	FLG-CAL-001	Design review (BOM, PRC), TRL 3	Met by design
UR-008	PR-012	Installation	FLG-CAL-001, FLG-BLD-001	Installation sequence review (CAL, BLD), TRL 3	Met on paper for the pilot route: 120 min, with no civil work (DDR-002)
UR-014	PR-013	Level datum	FLG-CAL-001, FLG-BLD-001	Survey method (CAL), TRL 3	Met on paper: tape survey plus a recorded dry-road radar reading
UR-006, UR-007	PR-014	Operating environment	FLG-CAL-001	Material and cell review (CAL), TRL 3	At risk:** inherits FieldNode's interior heat finding (FND R2, R3); the drain head module is rated -15 to 60 °C
UR-009, UR-010	PR-015	Mounting and mass on the pole	FLG-CAL-001, FLG-BLD-001, FLG-STM-001	Mass from the CAD model (CAL), TRL 3; weigh at TRL 4	At risk:** 4.994 kg (margin 0.006 kg, inside the accuracy of an estimate from model volumes); twist factor 1.8 with the M8 A4-80 anti-rotation through-bolt, 1.45 on friction alone on a 60 mm pole and 0.97 on a 40 mm pole
UR-016	PR-016	Cost	FLG-CAL-001, BOM	Priced BOM (bom/bom.csv), TRL 3	Over target by $45.50: $205.50 for the constructable design ($344.50 with the $139.00 FieldNode core)
UR-015	PR-017	Open data	FLG-CAL-001, FLG-BLD-001	Design review (PRC), TRL 3	Met by design
UR-017	PR-018	Tamper resistance	FLG-CAL-001	CAD model and BOM review (CAL, BLD), TRL 3	Met by design (new in this version; the fixings were added in FLG-DDR-003)
UR-010	PR-019	Cover profile and edges	FLG-CAL-001	CAD model (cad_check edge breaks) and drawing FLG-DWG-001, TRL 3	Met by design: hat section 16 mm high with 2 mm sheet; edges broken in the model

Gaps

None.